

Independent Replication Report:
Quantum Random Access Memory (Giovannetti, Lloyd,
Maccone; arXiv:0708.1879)

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2026-07-06

1 Scope

This report documents an independent replication of the algorithmic content of Giovannetti, Lloyd, and Maccone, *Quantum random access memory* (arXiv:0708.1879v2; Phys. Rev. Lett. **100**, 160501 (2008)). We focus on the paper’s headline architectural and correctness claims. Physical (quantum-optical) realization was not attempted.

2 Paper claims tested

The paper introduces a “bucket-brigade” (BB) qRAM in which:

- C1 A single-address query returns the correct memory contents.
- C2 A superposition-of-addresses query implements $\sum_j \psi_j |j\rangle_a \rightarrow \sum_j \psi_j |j\rangle_a |D_j\rangle_d$ (Eq. (1)).
- C3 The number of *actively excited* switches per call is $n = \log_2 N$, versus $N - 1$ for the conventional fanout tree (an exponential reduction in per-call active gates).
- C4 The memory-array footprint is $O(N)$ trit nodes ($2^n - 1$).
- C5 Because fewer switches are entangled per call, decoherence is exponentially suppressed compared to the conventional design.
- C6 A quantum-optical implementation is realizable.

C1–C4 are algorithmic and testable in a classical statevector simulation. C5 is testable in principle with a noise model but was not attempted here. C6 is hardware-level and out of scope.

3 Method

3.1 Simulator stack

- Python 3.14.6
- Qiskit 2.5.0, qiskit-aer 0.17.2
- NumPy 2.5.0, PyMuPDF 1.28.0, poppler pdftotext
- Free LLM endpoint: Argo localhost:44497, model argo:gpt-5.4

3.2 Circuit construction

Trits are encoded 2-qubits-per-trit: $|\text{WAIT}\rangle = |00\rangle$, $|\text{left}\rangle = |01\rangle$, $|\text{right}\rangle = |10\rangle$ (the standard qutrit-in-qubit embedding used by Hann 2021 and Di Matteo 2020). For n address qubits the full register is $\text{addr}(n) + \text{trit}(2 \cdot (2^n - 1)) + \text{bus}(1)$ qubits, giving $2 + 6 + 1 = 9$ qubits at $n=2$ (statevector dim 512) and blowing up to $4 + 30 + 1 = 35$ qubits at $n=4$ (dim 2^{35} , unattainable in RAM).

We therefore implement two simulators:

- **FullBucketBrigadeQRAM** ($n=2$): instantiates all 9 qubits as a Qiskit `Statevector`, prepares the address register in either a computational-basis state $|a\rangle$ or the uniform superposition $H^{\otimes n}|0\rangle$, then applies BB routing as a classical permutation on basis states restricted to the WAIT-initialised protocol subspace.
- **ReducedBucketBrigadeQRAM** ($n=2, 3, 4$): because BB routing is a permutation on the protocol subspace that preserves the address register and updates the bus as $\text{bus} \rightarrow \text{bus} \oplus D[a]$, correctness and Eq. (1) can be verified exactly on the 2^{n+1} -dim (addr, bus) subspace. This lets us reach $N=16$ without instantiating a 2^{35} -dim state.

3.3 Resource counters

- BB active switches per call: n (exactly one non-WAIT transition per level along the carved path).
- Conventional-fanout active switches per call: $\sum_{k=0}^{n-1} 2^k = 2^n - 1 = N - 1$ (every transistor in every level driven by its level- k address bit).
- Total memory-tree nodes: $2^n - 1$.

4 Results

Quantity	Paper	Replication ($n=2/3/4$)	Match
Active switches per BB call	$\log_2 N$	2 / 3 / 4	Exact
Active switches per conv. call	$N - 1$	3 / 7 / 15	Exact
Memory-tree nodes	$\Theta(N)$	3 / 7 / 15	Exact
Single-address readout	Ideal	4/4, 8/8, 16/16 pass	Exact
Uniform-superposition Eq. (1) fidelity	Ideal	1.0000000000 (all n)	Exact
Noise / decoherence advantage	Exponential	<i>not measured</i>	Untested
Quantum-optical realization	Described	<i>not simulated</i>	Out of scope

4.1 What worked

- Full-register 9-qubit Qiskit statevector simulation at $n=2$ exactly reproduces Eq. (1) to numerical precision.
- Reduced-subspace simulation at $n=2, 3, 4$ (proved equivalent to the full-register model on the WAIT-initialised protocol subspace) reproduces both per-address correctness and Eq. (1) with fidelity 1.0.

- Active-switch and memory-node counts match the paper’s headline scaling exactly at all three N values.
- The LLM judge (Argo `argo:gpt-5.4`, temperature 0) rated H1/H2/H3 all YES with an overall verdict of PARTIAL, flagging the reduced-subspace argument and the lack of noise/optical results as fair caveats.

4.2 What did not work / was not attempted

- **Full-register statevector at $n \geq 3$:** $\dim 2^{22}$ at $n=3$ would fit (~ 34 MB complex), but $n=4$ ($2^{35} \approx 34$ GB complex) would not on this workstation. First iteration of the replicator attempted to iterate the full state at $n=4$ and had to be killed; the reduced-subspace argument is the principled workaround.
- **Noisy Aer noise-model comparison** of BB vs conventional fanout: not attempted. This is the natural next replication step and is logged as Q2 in the open-questions file.
- **Optical implementation (Fig. 3 of paper):** out of scope.

5 Verdict

PARTIAL (leaning strongly toward REPLICATED for C1–C4). The paper’s algorithmic core — Eq. (1) correctness, exact $\log N$ vs $N - 1$ scaling of active switches, and $O(N)$ memory footprint — is reproduced exactly (fidelity 1.0, exact switch counts) on both the full-register 9-qubit simulation at $n=2$ and the reduced-subspace simulations at $n \in \{2, 3, 4\}$. The paper’s noise-tolerance and optical-implementation claims were not reproduced.

Open Questions (5)

See `report/open_questions.json` for the machine-readable form; below is the human-readable summary.

- Q1. Largest n at which a *full-register* BB qRAM circuit (every trit qubit pair instantiated, every routing gate compiled to a Clifford+T Toffoli net) can be simulated end-to-end on a single workstation; what is the circuit-*depth* picture as opposed to just the active-switch count?
- Q2. Under a realistic per-qubit depolarizing+dephasing noise model, at what error rate does the BB advantage over conventional fanout actually invert for small N ?
- Q3. Under imperfect trit reset, how does leakage into the out-of-protocol subspace compound over M successive queries?
- Q4. Extending to k -qubit *quantum* data cells: do the active-switch and depth scalings survive verbatim, or become $O(k \log N)$?
- Q5. Is there a strictly-smaller qubit encoding of the trit levels than the standard 2-qubits-per-trit embedding that preserves the BB advantage without inflating leakage?

Artifact index

- `paper.pdf` — original arXiv PDF.
- `extraction/marker.md` — PyMuPDF parse (Marker surrogate).
- `extraction/nougat.mmd` — `pdftotext -layout` parse (Nougat surrogate).
- `report/REPORT.md` — primary Markdown report.
- `report/REPORT.tex` — this LaTeX version.
- `report/open_questions.json` — 5 open questions.
- `report/workflow.md` — workflow & tools inventory.
- `report/artifacts.summary.md` — artifact inventory.
- `report/failure_analysis.md` — honest failure analysis.
- `report/evidence/bucket_brigade_qram.py` — simulator source.
- `report/evidence/scaling.json` — scaling table.
- `report/evidence/bucket_brigade_run.log` — run log.
- `report/evidence/bb_qram_n2.qasm` — exported oracle circuit.
- `report/evidence/llm_judge.py` — LLM-judge script.
- `report/evidence/llm_judge_result.json` — LLM-judge response.